

Data Collection and Preprocessing Phase

Date	26 July 2026
Team ID	MIRUTHALA M K
Project Title	Global AI Adoption In Education
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	A structured Global AI Adoption in Education dataset containing country, region, year, student AI usage, teacher AI usage, rural AI usage, urban AI usage, AI tool usage, and AI adoption level information. The dataset is used to analyze global AI adoption patterns and create interactive Tableau visualizations.
Data Cleaning	Missing Values: The dataset was checked for missing or incomplete values before analysis. Duplicates: Duplicate records were reviewed and removed where necessary to maintain data integrity. Error Correction: Inconsistent country, region, category, and text values were checked and standardized to ensure proper grouping and filtering in Tableau.
Data Transformation	Calculated Fields: KPI measures such as average daily AI usage, average rural AI usage, and average urban AI usage were used for analysis. Aggregations: Data was aggregated to calculate country-level AI usage, student and teacher usage, year-wise adoption trends, regional adoption, and AI tool market share. Filtering: Data was filtered based on countries, years, regions, AI tools, and adoption levels to support different visualizations.
Data Type Conversion	Country, region, and AI tool fields were treated as text or categorical data. Year was treated as a numerical/time-based field. AI usage percentages, adoption percentages, and daily

	usage values were treated as numerical measures to support calculations and visualizations in Tableau.
Column Splitting and Merging	Data fields were organized into separate logical columns such as country, region, year, student usage, teacher usage, rural usage, urban usage, AI tool, and adoption level. This structure enables responsive filtering and comparison across different dimensions in Tableau.
Data Modeling	The dataset was structured to support analysis across multiple dimensions, including country, region, year, user group, community type, AI tool, and adoption level. These dimensions were used to create relationships and comparisons within Tableau visualizations.
Save Processed Data	After data preparation and validation, the structured dataset was used as the data source for Tableau. The cleaned and prepared data was connected to Tableau to create KPI cards, maps, bar charts, trend charts, pie charts, and comparative visualizations.